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ADDITIONAL MATHEMATICS

Paper 4037/11

Paper 11

Key message

In order to ensure that candidates receive credit where possible for incorrect answers, they should remember to show all their working. Incorrect answers without working cannot be given any credit, whereas partial credit can be awarded if a correct method is shown.

General comments

The majority of candidates were sufficiently well prepared to make a reasonable attempt at most of the questions on the paper. Some candidates would have benefited from more practice in most topic areas.

As noted above, candidates should be reminded of the need to show all the steps in their working, particularly in questions where the answer is given, such as **Qs 1**, **6** and **12 OR**.

In order not to lose marks unnecessarily, candidates need to give non-exact numerical answers to 3 significant figures, which should require working to at least 4 significant figures during the solution of a question.

Comments on specific questions

Question 1

There were many excellent solutions starting from the left-hand side of the identity. Those starting from the right-hand side made little useful progress. Candidates need to understand that all significant steps in the argument should be shown in order to gain maximum credit.

Question 2

Few candidates managed to get full marks on this question. Many candidates obtained $\log ab^3$ but could not incorporate the 3 into a single logarithm.

Answer: $\log\left(\frac{ab^3}{1000}\right)$

Question 3

(a) Most candidates were able to identify and shade the required regions.

(b) Many candidates did not appreciate the difference between P and n(P).

Answer: **(b)** 3

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Question 4

(a) Most candidates attempted the simplest method of solution which was to convert both t into powers of 2. Those who attempted to convert into powers of 4, 8, or 16 were muc successful. A few candidates were able to obtain the solution by taking logs.

(b) This question was generally done well, and most candidates were able to use the laws of indices to gain full marks.

Answers: **(a)** $\frac{4}{3}$ **(b)** $p = 1, \quad q = -\frac{4}{5}$

Question 5

(i) The majority of candidates realised that a sine curve between –2 and 4 was required. Many slips were made with regard to starting and finishing at y = 1 which resulted in there not being exactly one cycle of the graph.

(ii) The concept of modulus did not seem to be generally understood, and so this is an area where more practice would be beneficial.

(iii) Most candidates who had answered part **(ii)** correctly were able to relate the answer to this question to their graph in part **(ii)**. It was common to see attempts to list the solutions of the equation, rather than write down the number of solutions. Candidates should ensure that they read the question carefully to ascertain what is required.

Answer: **(iii)** 5

Question 6

(i) Most candidates solved the equations simultaneously leading to x = 2. A few correct solutions were also obtained by substituting x = 2 in both equations, leading to y = 4.

(ii) Most candidates differentiated successfully and substituted x = 2 to obtain the required result.

(iii) This part was generally well done by candidates.

Answer: **(iii)** $y = 4x - 4$

Question 7

Many candidates took the opportunity to present a well set out, step-by-step solution to this problem, and gained full marks. Any mistakes made were usually the result of not reading the question carefully enough, and a number of candidates found either the equation of the perpendicular through *B*, or through the mid-point of *AB*, or took *C* as a point on the *x*-axis.

Answer: 12.5

Question 8

(a) As well as the two correct matrix products, other incorrect ones were often included. A number of candidates wasted time by evaluating the matrix products despite the instruction in the question not to do so.

(b) Most candidates knew how to find $\mathbf{X}^{-1}$ and substitute it in the given equation, and this was the most popular method of solution. The alternative of pre-multiplying the equation by **X** was only occasionally seen.

Answer: $x = 4 \quad y = 12$

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Question 9

(i) This part question was answered well, although $v = 0$ was a regularly occurring incorrect an

(ii) Most candidates realised the need to differentiate to find the acceleration and reached $\sin^{-1}(-0$ A substantial number of candidates then lost marks by using degrees rather than radians.

(iii) The need for integration was generally appreciated, but again marks were lost due to the use of degrees rather than radians.

Answers: **(i)** 5 **(ii)** 0.916 **(iii)** 1.14

Question 10

(a) (i) Those candidates who realised the need for substitution found this an easy mark.

(ii) The integration was usually done well. A number of candidates omitted the + *c*. A few chose, incorrectly, to propose a straight line equation for the curve, using the given derivative as the gradient.

(b) (i) This was generally well done. There were a number of incorrect attempts which involved integrating the term inside the bracket to $\frac{7}{2}x^2 + 8x$.

(ii) The majority of candidates knew how to use limits. A few missed the lower limit, assuming that it would come to zero. As this was a "Hence" question, correct answers from a calculator following incorrect working were given no credit.

Answers: **(a) (i)** 5 **(ii)** $y = -5e^{1-x} - x^3 + 10$ **(b) (i)** $\frac{3}{28}(7x+8)^{\frac{4}{3}}$ **(ii)** 25.7

Question 11

(a) (i) A good number of candidates were able to complete the square successfully. Others expanded f(x) and equated coefficients.

(ii) There were many possible correct domains which candidates could, and did, provide.

(b) (i) Many candidates knew how to find the range of g. Very few realised that the domain of h was the range of h^{-1}.

(ii) Most candidates could sketch the graph of $y = g(x)$ as a parabola starting at 4 on the y-axis. Far fewer were able to sketch the graph of $y = g^{-1}(x)$ using the symmetry about the line $y = x$.

(iii) The majority of candidates got the order of operations correct and most were able to obtain a quadratic equation which they solved. Very few realised that the solution $x = 4$ was inadmissible, as h(4) is negative, and therefore not in the domain of g.

Answers: **(a)(i)** $2(x-2)^2 - 3$ **(ii)** $x \geq 2$ **(b)(i)** $g(x) \geq 4$ $h^{-1}(x) \geq 0$ **(iii)** 8.5

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Question 12 EITHER

This alternative was attempted by the majority of candidates. It is possible that some were tempted by the relatively straightforward part **(i)** as many of them gained full marks in part **(i)** and were then unable to make a worthwhile attempt at parts **(ii)** and **(iii)**.

(i) Many candidates found this an easy 6 marks. $\frac{\mathrm{d}y}{\mathrm{d}x}$ was found either by the product rule or, more frequently, by expansion. Candidates should note that, when determining the nature of the stationary points, it is not sufficient to write down the answers without showing any working or explanation.

(ii) Most candidates realised that they should be using the chain rule and that $\frac{\mathrm{d}z}{\mathrm{d}t} = 10$, but few were able to obtain a correct value for $\frac{\mathrm{d}y}{\mathrm{d}t}$. In many cases the notation used was poor, and often confused with the small changes formula.

(iii) Although, in general, this part was also poorly done, more candidates realised that they needed to divide their answer to part **(ii)** by $\frac{\mathrm{d}y}{\mathrm{d}x}$.

Answers: **(i)** $x = \frac{2}{3}$ maximum, $x = 4$ minimum **(ii)** $-\frac{5}{6}$ **(iii)** $\frac{5}{48}$

Question 12 OR

This alternative proved to be a much less popular choice but, on the whole, candidates performed better on it than on "Either".

(i) Most candidates showed ample working and recognised the need to produce clear evidence of simplification to produce the given result.

(ii) The differentiation was generally well done. A few candidates stopped at x = 3 and did not give the dimensions of the cuboid.

(iii) Most candidates were familiar with the procedures for evaluating small changes and interpreted their answer correctly.

Answers: **(ii)** 3 by 6 by 4 **(iii)** –38p decrease

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ADDITIONAL MATHEMATICS

Paper 4037/12

Paper 12

Key message

In order to ensure that candidates receive credit where possible for incorrect answers, they should remember to show all their working. Incorrect answers without working cannot be given any credit, whereas partial credit can be awarded if a correct method is shown.

General comments

In general, most candidates appear to have been well prepared for this paper and attempted all the questions. There were many high scoring scripts showing that candidates had a thorough knowledge of the syllabus and were able to apply techniques learned both appropriately and correctly in the great majority of cases.

The new format of the paper was helpful to candidates as they were encouraged by the constraints of space to set out their work tidily and with greater thought. Errors in reading information from the questions were far fewer than in previous sessions.

Centres should note that additional sheets of paper, including graph paper, should only be issued if a candidate runs out of space for a particular question. Where candidates are required to draw an accurate graph, a grid is printed on the question paper.

Comments on specific questions

Question 1

Most candidates attempted to use a method using the discriminant of the quadratic equation formed when y was equated to zero. Errors occurred in the substitution of the terms involving k, but most candidates were able to achieve some measure of success.

Alternative methods involving the gradient function were usually less successful due to more involved algebraic manipulation. Candidates then tended to forget the actual aim of the question.

Answer: $k = -2$

Question 2

The required coefficients were usually identified correctly; however, errors were often made when dealing with the powers of a and $\frac{a}{3}$, and also with the relationship between the two coefficients. As a result, completely correct solutions were unusual although candidates were able to gain most of the marks available for using a correct method.

Answer: $a = \frac{1}{6}$

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Question 3

(a) Most candidates were able to identify at least one of the unknowns from the given graph and the correct numerical value required. Errors in identifying m and p were the most common, with candidates getting them the wrong way round.

(b)(i) Most candidates were able to obtain the correct amplitude.

(ii) Most candidates were able to obtain the correct period, although some chose to give the period in degrees rather than radians in terms of π, as asked for in the question.

Answers: **(a)** $k = 2,\ m = 3,\ p = 1$, **(b)(i)** 5, **(ii)** $\frac{2\pi}{3}$

Question 4

Candidates were instructed not to use a calculator for this question. The intention in such questions is for candidates to show all their workings thus demonstrating, in this case, their knowledge of how to deal with surds. Some evidence of expansion of brackets containing surds was required for each part of this question.

(i) A correct trigonometric equation was obtained initially in most cases, with candidates realising the need for rationalisation. The majority of candidates gave a completely correct solution.

(ii) The area of the triangle was usually obtained correctly in an unsimplified form. Some errors in cancelling the 0.5 with single terms within one of the brackets were unfortunately too common.

(iii) Most candidates realised that they needed to work out AC^2. However, many did not realise that this gave the value of the area required and continued with further, unnecessary, calculations.

Answers: **(i)** $2\sqrt{2}$, **(ii)** $4 + 3\sqrt{2}$, **(iii)** $27 + 18\sqrt{2}$

Question 5

(i) This part was very well done by the great majority of candidates; the most usual method involved showing $\mathrm{f}(0.5) = 0$. Methods involving long division were also used, usually correctly. Those candidates that used this approach had then done most of the work required for part **(ii)**.

(ii) Most candidates realised that a quadratic factor needed to be found. This was usually done by algebraic long division, but many adopted a method comparing coefficients and some were able to do the factorisation by observation. Most candidates were able to gain the first two marks available, but there was a great deal of misinterpretation of what was required after this. It was hoped that candidates would look at the discriminant of their quadratic factor (or equate the quadratic factor to zero and attempt to solve) and realise that there were no real roots, so the only real root to the solution of the initial cubic equation was 0.5. Many candidates attempted to factorise the quadratic factor arriving with real solutions. For those candidates that did use a correct approach, the statement of the value of the real root was often omitted.

Answer: 0.5

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Question 6

(i) An equation of a straight line involving $\lg y$ and x was produced by most candidates, with the occasional error in the calculation of the constant involved.

(ii) Correct solutions for this part were rare. Candidates found it difficult to rearrange their answer to part **(i)** in terms of powers of ten. Some candidates were able to determine the correct value of b, but were unable to express a in a correct form.

Answers: **(i)** $\lg y = \frac{1}{5}x + 2$, **(ii)** $a = 100$, $b = \frac{1}{5}$

Question 7

(i) Most candidates obtained a correct answer.

(ii) Apart from those candidates who mistakenly used permutations rather than combinations, most realised that they needed the product of two combinations and were able to obtain a correct answer.

(iii) A common response was 336, with candidates having forgotten the case where there were no women on the team. Otherwise, as before, this part was usually done well.

Answers: **(i)** 3003, **(ii)** 1050, **(iii)** 364

Question 8

(i) Most candidates produced a correct sketch in the space given on the question paper. However, many did not state all of the coordinates of the points where the curve meets the coordinate axes as requested in the question. They were often not even marked on their sketch, which would have been acceptable. Some Centres issued graph paper for this question which was not necessary.

(ii) A surprisingly high number of candidates could not state the coordinates of the stationary point correctly.

(iii) Correct graphs showing the modulus function were common, although many candidates attempted to draw a 'modulus' graph using the line $y = 5$.

Answer: **(ii)** (1, –9)

Question 9

(i) This part of the question proved to be quite difficult for some candidates. They were required to use the properties of an isosceles triangle in order to gain the required result. Many attempts using the arc length were seen but these were ineffective.

(ii) The value of the radius found using the arc length was usually done correctly. Common errors included omitting π from the arc length 9π together with incorrect arithmetic.

(iii) Correct methods were employed by most candidates with the occasional error in calculation. Those that did not use the sine rule for the area of a triangle, but chose instead to find the length AB and use $\frac{1}{2} \times$ base $\times$ height, appeared to be more likely to make errors in calculation and use figures that had been rounded prematurely, thus resulting in an inaccurate value for the area.

Answers: **(ii)** 15, **(iii)** 105

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Question 10

(i) Whilst most candidates were able to find $\overrightarrow{AB}$ correctly, there were many who did not und the term unit vector. Many candidates did attempt to find the magnitude of their vector but unsure as to what to do with this magnitude. Some candidates used incorrect notation. Centr should encourage candidates to be aware of the correct notation to use when dealing with vectors. Column vector notation and **i**~**j** notation is acceptable. Candidates should be discouraged from writing their vectors in a form that is similar to that used for coordinates.

(ii) More success was to be had by most in this part, with many applying a correct approach (of which there were many) to finding the position vector of C. The same comments regarding notation mentioned in part **(i)** also apply to this part of the question.

Answers: **(i)** $\frac{1}{25}\begin{pmatrix}24\\-7\end{pmatrix}$, **(ii)** $\begin{pmatrix}41\\-16.5\end{pmatrix}$

Question 11

Candidates should be reminded that angles in degrees should always be given correct to one decimal place, whilst angles in radians should be given correct to 3 significant figures.

(i) Provided candidates used either the correct trigonometric identity and obtained a quadratic equation in terms of cosec x, or attempted to obtain a quadratic equation in terms of $\sin x$, a successful solution was usually obtained. Most errors occurred with the substitution of an incorrect identity, together with poor manipulation of the trigonometric terms involved. Candidates should be reminded that these identities are given on the Mathematical Formulae page at the front of the paper.

(ii) Completely correct solutions were common, with most candidates recognising that the equation had to be written in terms of $\tan 2y$. Some candidates failed to deal with the double angle.

(iii) It was pleasing to see many correct solutions for this question. Questions of this type often prove difficult for candidates, as they have to deal with both the correct order of operations and the fact that radians are involved. There were of course exceptions, with some candidates using a mixture of both radians and degrees and some who dealt with $\cos\left(z+\frac{\pi}{6}\right)$ as a sum of two cosines.

Answers: **(i)** $19.5^\circ, 160.5^\circ$, **(ii)** $25.7^\circ, 115.7^\circ$, **(iii)** $\frac{\pi}{2}, \frac{7\pi}{6}$

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Question 12 Either

Many candidates either misunderstood what was required of them, or misinterpreted the instruc information given. They should be reminded to read the question carefully. A sketch of the situation by the candidates themselves would have probably helped them in a question that was written to test skills of taking the information given and determining a correct approach to adopt in order to achieve the required solution. Those candidates that did draw a sketch were usually able to gain full marks for this question.

(i) Many candidates did not attempt to find the tangent to the curve and invariably gave the coordinates of the point A as $(-1, 5)$. A sketch at this point would have helped most candidates.

(ii) Most candidates were able to state the coordinates of the point B correctly and go on to find the equation of the normal at B and make an attempt at finding the coordinates of the point D. Some correct methods to find the area of the triangle formed were seen. Candidates were usually more successful if they used the matrix approach rather than $\frac{1}{2} \times$ base $\times$ height. Again, a sketch would have helped many to sort out which point was which and what the area they were finding actually looked like.

Answers: **(i)** (0, 5) **(ii)** 50

Question 12 Or

This question was also written in order to test the candidates' ability to sort through information given and formulate a plan in order to obtain the required result. Provided a careful and logical approach was adopted, many candidates were able to gain full marks for this unstructured question.

Most candidates were able to find the coordinates of the points P, Q and R successfully using differentiation together with the fact that the given tangent is parallel to the x-axis. Integration was then required, a fact that most candidates recognised and, apart from the occasional error, this was usually done successfully with the correct area being identified and calculated.

Answer: 4

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$\mathscr{E}$)=72 **(ii)** $R \subset W$ or $R \cap W = R$ or $R \cup W = W$ or $R \cap W' = \varnothing$ or others

(b)(ii) $(X' \cap Y)'$ or $(X \cap Y) \cup Y'$ or $(X \cup Y)' \cup X$ or $(X' \cap Y') \cup X$ or others

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